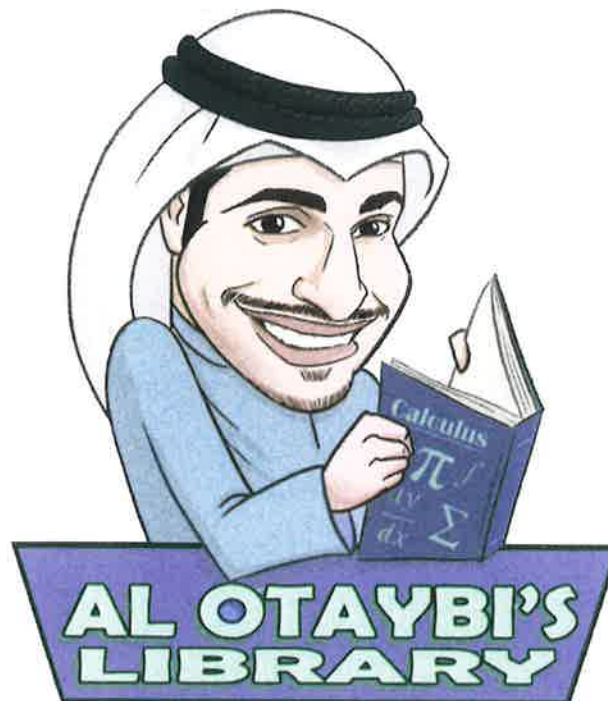


Kuwait University

2021/2022

Fall course



Calculus A

First

Section : 1.5 (part b)

1.5 Inverse Functions and Logarithms

(4)



Text book questions



Find a formula for the inverse of the function

$$f(x) = e^{2x-1}$$

$$\begin{aligned} y &= e^{2x-1} \\ \ln y &= \ln e^{2x-1} \\ \ln y &= 2x-1 \\ \ln y + 1 &= 2x \end{aligned} \quad \left| \begin{array}{l} \xrightarrow{\div 2} \\ \therefore f^{-1}(x) = \frac{\ln x + 1}{2} \end{array} \right.$$

Find a formula for the inverse of the function

$$f(x) = \ln(x+3)$$

$$\begin{aligned} y &= \ln(x+3) \\ e^y &= e^{\ln(x+3)} \\ e^y &= x+3 \end{aligned} \quad \left| \begin{array}{l} \xrightarrow{\quad} \\ \therefore f^{-1}(x) = e^x - 3 \end{array} \right.$$

Find a formula for the inverse of the function

$$f(x) = \frac{1 - e^{-x}}{1 + e^{-x}}$$

$$\begin{aligned} y &= \frac{1 - e^{-x}}{1 + e^{-x}} \\ y(1 + e^{-x}) &= 1 - e^{-x} \\ y + ye^{-x} &= 1 - e^{-x} \\ e^{-x} + ye^{-x} &= 1 - y \\ e^{-x}(1 + y) &= 1 - y \\ \div (1 + y) & \end{aligned} \quad \left| \begin{array}{l} \xrightarrow{\quad} \\ e^{-x} = \frac{1-y}{1+y} \\ \ln e^{-x} = \ln \frac{1-y}{1+y} \\ -x = \ln \frac{1-y}{1+y} \\ * (-1) \\ x = -\ln \frac{1-y}{1+y} \\ \therefore f^{-1}(x) = -\ln \frac{1-x}{1+x} \end{array} \right.$$

Let $f(x) = e^{x+1} + 2$. Find $f^{-1}(x)$

July 2017

$$y = e^{x+1} + 2$$

$$y - 2 = e^{x+1}$$

$$\ln(y-2) = \ln e^{x+1}$$

$$x+1 = \ln(y-2)$$

$$x = \ln(y-2) - 1$$

$$\Rightarrow f^{-1}(x) = \ln(x-2) - 1$$

Let $f(x) = \ln\left(\frac{x}{2x-1}\right)$. Find $f^{-1}(x)$

Nov 2017

$$y = \ln\left(\frac{x}{2x-1}\right)$$

$$e^y = e^{\ln\left(\frac{x}{2x-1}\right)}$$

$$e^y = \frac{x}{2x-1}$$

$$e^y(2x-1) = x$$

$$2xe^y - e^y = x$$

$$2xe^y - x = e^y$$

$$x[2e^y - 1] = e^y$$

$$\div [2e^y - 1]$$

$$x = \frac{e^y}{2e^y - 1}$$

$$\therefore f^{-1}(x) = \frac{e^x}{2e^x - 1}$$



Tricky question !!



Let $f(x) = \ln \left[\frac{1+e^x}{2+e^x} \right]$ Find $f^{-1}(x)$

$$y = \ln \left[\frac{1+e^x}{2+e^x} \right]$$

$$e^y = e^{\ln \left[\frac{1+e^x}{2+e^x} \right]}$$

$$e^y = \frac{1+e^x}{2+e^x}$$

$$e^y [2+e^x] = 1+e^x$$

$$2e^y + e^x e^y = 1+e^x$$

$$2e^y - 1 = e^x - e^x e^y$$

$$2e^y - 1 = e^x (1 - e^y)$$

$$e^x (1 - e^y) = 2e^y - 1$$

$$\div (1 - e^y)$$

$$e^x = \frac{2e^y - 1}{1 - e^y}$$

$$\ln e^x = \ln \frac{2e^y - 1}{1 - e^y}$$

$$x = \ln \frac{2e^y - 1}{1 - e^y} \Rightarrow f^{-1}(x) = \ln \frac{2e^x - 1}{1 - e^x}$$

Let $f(x) = \ln(2x + 3)$

a) Find $f^{-1}(x)$

b) Find the domain and the range of the function f

Nov 2016

a) $y = \ln(2x + 3)$

$$e^y = e^{\ln(2x+3)}$$

$$e^y = 2x + 3$$

$$e^y - 3 = 2x$$

$$\therefore x = \frac{e^y - 3}{2}$$

b) domain of f :

$$2x + 3 > 0$$
$$2x > -3$$
$$x > -\frac{3}{2}$$

$$\Rightarrow \therefore D_f = \left(-\frac{3}{2}, \infty\right)$$

domain of $f^{-1} = \mathbb{R}$

$$\Rightarrow \therefore \text{range of } f = \mathbb{R}$$

Let $f(x) = 1 + \ln(x + 3)$

a) Find the domain of f

b) Find $f^{-1}(x)$

c) Find the range of f

Mar 2015

a) $x + 3 > 0$

$x > -3 \longrightarrow \therefore \text{domain of } f = (-3, \infty)$

b) $y = 1 + \ln(x + 3)$

$y - 1 = \ln(x + 3)$

$e^{y-1} = e^{\ln(x+3)}$

$e^{y-1} = x + 3$

$x = e^{y-1} - 3$

$\Rightarrow \therefore f^{-1}(x) = e^{x-1} - 3$

c) domain of $f^{-1}(x) = \mathbb{R}$

$\therefore \text{range of } f^{-1}(x) = \mathbb{R}$

Let $f(x) = 1 + e^{(x+1)}$

a) Find the domain of f

b) Find $f^{-1}(x)$

c) Find the range of f

June 2015

a) Domain of $f = \mathbb{R}$

b) $y = 1 + e^{x+1}$

$e^{x+1} = y - 1$

~~$\ln e^{x+1} = \ln(y-1)$~~

$x+1 = \ln(y-1)$

$x = \ln(y-1) - 1$

$\Rightarrow \therefore f^{-1}(x) = \ln(x-1) - 1$

c) range of f :

domain of $f^{-1}(x)$: $x-1 > 0$
 $x > 1$
 $(1, \infty)$

$\Rightarrow \therefore \text{Domain of } f^{-1} = (1, \infty)$

$\therefore \text{range of } f = (1, \infty)$

Let $f(x) = 3^x + 1$.

a) Find $f^{-1}(x)$

b) Find the range of f

April 2018

a) $y = 3^x + 1$

$$y - 1 = 3^x$$

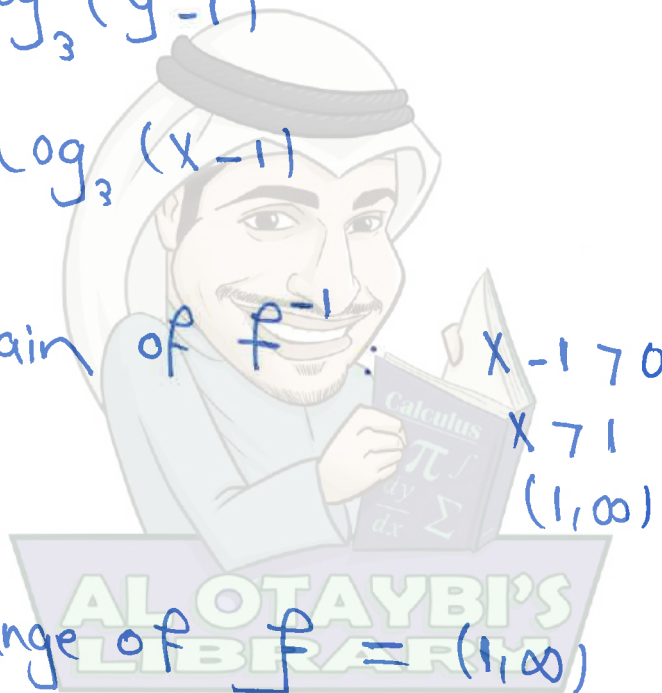
$$\log_3(y-1) = \log_3 3^x$$

$$x = \log_3(y-1)$$

$$f^{-1}(x) = \log_3(x-1)$$

b) Domain of f^{-1} : $x-1 \geq 0$
 $x \geq 1$
 $(1, \infty)$

$\Rightarrow$ $\therefore$ Range of $f = (1, \infty)$



$$\text{Let } f(x) = \ln\left(\frac{2}{x} + 3\right)$$

a) Find the inverse function of f

b) Find the range of f

Oct 2018

$$a) \quad y = \ln\left(\frac{2}{x} + 3\right)$$

$$e^y = e^{\ln\left(\frac{2}{x} + 3\right)}$$

$$e^y = \frac{2}{x} + 3$$

$$\ast x \quad e^y - 3 = \frac{2}{x}$$

$$\div (e^y - 3) \quad (e^y - 3)x = 2$$

$$x = \frac{2}{e^y - 3}$$

$$\therefore f^{-1}(x) = \frac{2}{e^x - 3}$$

$$b) \quad e^x - 3 = 0$$

$$e^x = 3$$

$$\ln e^x = \ln 3$$

$$x = \ln 3$$

$$\therefore \text{Domain of } f^{-1}(x) = \mathbb{R} \setminus \{\ln 3\}$$

$$\therefore \text{Range of } f(x) = \mathbb{R} \setminus \{\ln 3\}$$

Let $f(x) = \ln(2^x - 1)$. Find $f^{-1}(x)$

Mar 2018

$$y = \ln(2^x - 1)$$

$$e^y = e^{\ln(2^x - 1)}$$

$$e^y = 2^x - 1$$

$$e^y + 1 = 2^x$$

$$\log_2(e^y + 1) = \log_2 2^x$$

$$x = \log_2(e^y + 1)$$

$$\Rightarrow \therefore f^{-1}(x) = \log_2(e^x + 1)$$

Extra

b) Find the domain of $f(x)$.

c) Find the range of $f(x)$.

b) Domain of f :

$$2^x - 1 > 0$$

$$2^x > 1$$

$$\log_2 2^x > \log_2 1$$

$$x > 0$$

$$\Rightarrow \therefore D_f = (0, \infty)$$



c) Domain of f^{-1}

$$e^x + 1 > 0$$

$$e^x > -1$$

$\therefore$ Domain of $f^{-1} = \mathbb{R}$

$\therefore$ Range of f

$$\Rightarrow \therefore R_f = \mathbb{R}$$





Text book question



Let $f(x) = \ln(e^x - 3)$

a) Find the domain of $f(x)$

b) Find $f^{-1}(x)$

c) Find the range of $f(x)$

a) Domain of $f(x)$: $e^x - 3 > 0$

$$e^x > 3$$

$$\cancel{\ln e^x} > \ln 3$$

$$x > \ln 3$$

$$(\ln 3, \infty)$$

$$\Rightarrow D_f = (\ln 3, \infty)$$

b) $y = \ln(e^x - 3)$

$$e^y = \cancel{e^{\ln(e^x - 3)}}$$

$$e^y = e^x - 3$$

$$e^y + 3 = e^x$$

$$\ln(e^y + 3) = \cancel{\ln e^x}$$

$$x = \ln(e^y + 3)$$

$$\Rightarrow \therefore f^{-1}(x) = \ln(e^x + 3)$$

c) Domain of $f^{-1}(x)$: $e^x + 3 > 0$

$$e^x > -3$$

$$D_{f^{-1}} = \mathbb{R} \rightarrow \therefore \text{Range of } f = \mathbb{R}$$

Let $f(x) = \ln(2e^x - 1)$

a) Find the domain of $f(x)$

b) Find $f^{-1}(x)$

c) Find the range of $f(x)$

Oct 2015

a) Domain: $2e^x - 1 > 0$
 $2e^x > 1$
 $\div 2$ $e^x > \frac{1}{2}$
 $\ln e^x > \ln \frac{1}{2}$
 $x > \ln \frac{1}{2} \Rightarrow (\ln \frac{1}{2}, \infty)$

b) $y = \ln(2e^x - 1)$

$e^y = e^{\ln(2e^x - 1)}$

$e^y = 2e^x - 1$

$e^y + 1 = 2e^x$

$\div 2$

$e^x = \frac{e^y + 1}{2}$

$x = \ln \frac{e^y + 1}{2}$

$\Rightarrow f^{-1}(x) = \ln \frac{e^x + 1}{2}$

c) Domain of $f^{-1}(x)$: $\frac{e^x + 1}{2} > 0$

$\times 2$

$e^x + 1 > 0$

$e^x > -1$

$\Rightarrow \therefore \text{Domain of } f^{-1}(x) = \mathbb{R}$

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$\Rightarrow \therefore \text{range of } f(x) = \mathbb{R}$

$$\text{Let } f(x) = [\ln(x-1) + 1]^3$$

a) Find the domain of $f(x)$

b) Find $f^{-1}(x)$

c) Find the range of $f(x)$

a) Domain of $f(x)$: $x-1 > 0$
 $x > 1$
 $(1, \infty)$

$$\Rightarrow D_f = (1, \infty)$$

b) $y = [\ln(x-1) + 1]^3$

$$y^{\frac{1}{3}} = \ln(x-1) + 1$$

$$y^{\frac{1}{3}} - 1 = \ln(x-1)$$

$$e^{(y^{\frac{1}{3}} - 1)} = e^{\ln(x-1)}$$

$$e^{(y^{\frac{1}{3}} - 1)} = x - 1$$

$$x = e^{y^{\frac{1}{3}} - 1} + 1$$

$$\Rightarrow f^{-1}(x) = e^{(x^{\frac{1}{3}} - 1)} + 1$$

c) Domain of $f^{-1}(x) = \mathbb{R}$

$\therefore$ Range of $f(x) = \mathbb{R}$

$$\text{Let } f(x) = \frac{x+1}{2x+1}$$

a) Show that f is one-to-one.

b) Find the inverse function of f .

c) Find the domain and range of f .

Mar 2019

a) if

$$f(x_1) = f(x_2)$$

$$\frac{x_1+1}{2x_1+1} = \frac{x_2+1}{2x_2+1}$$

$$(x_1+1)(2x_2+1) = (x_2+1)(2x_1+1)$$

$$2x_1x_2 + x_1 + 2x_2 + 1 = 2x_1x_2 + x_2 + 2x_1 + 1$$

$$x_1 + 2x_2 = x_2 + 2x_1$$

$$2x_2 - x_2 = 2x_1 - x_1$$

$$\therefore x_2 = x_1$$

$\therefore f$ is one-to-one.

b)

$$y = \frac{x+1}{2x+1}$$

$$y(2x+1) = x+1$$

$$2xy + y = x + 1$$

$$2xy - x = 1 - y$$

$$x[2y-1] = 1-y$$

$$x = \frac{1-y}{2y-1}$$

$$f^{-1}(x) = \frac{1-x}{2x-1}$$

c) Domain of $f(x)$:

$$2x + 1 = 0$$

$$2x = -1 \longrightarrow x = -\frac{1}{2}$$

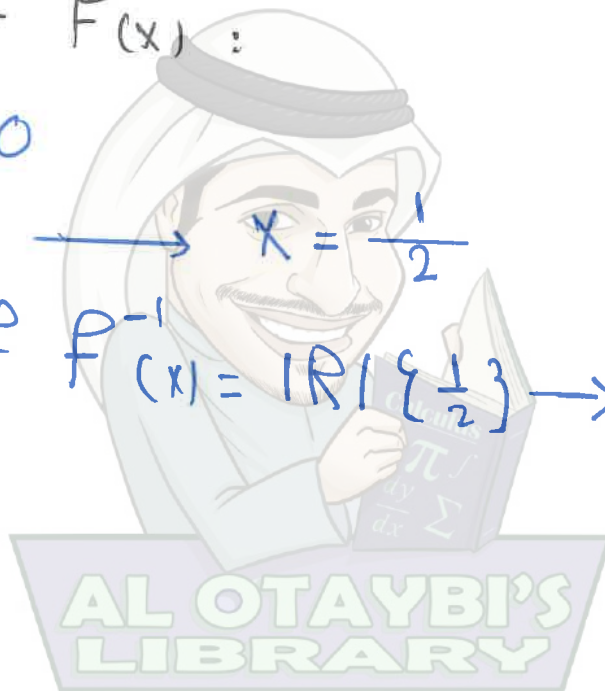
Domain of $f(x) = \mathbb{R} \setminus \{-\frac{1}{2}\} \longrightarrow$ Range of $f^{-1} = \mathbb{R} \setminus \{-\frac{1}{2}\}$

Domain of $f^{-1}(x)$:

$$2x - 1 = 0$$

$$2x = 1 \longrightarrow x = \frac{1}{2}$$

Domain of $f^{-1}(x) = \mathbb{R} \setminus \{\frac{1}{2}\} \longrightarrow$ Range of $f = \mathbb{R} \setminus \{\frac{1}{2}\}$





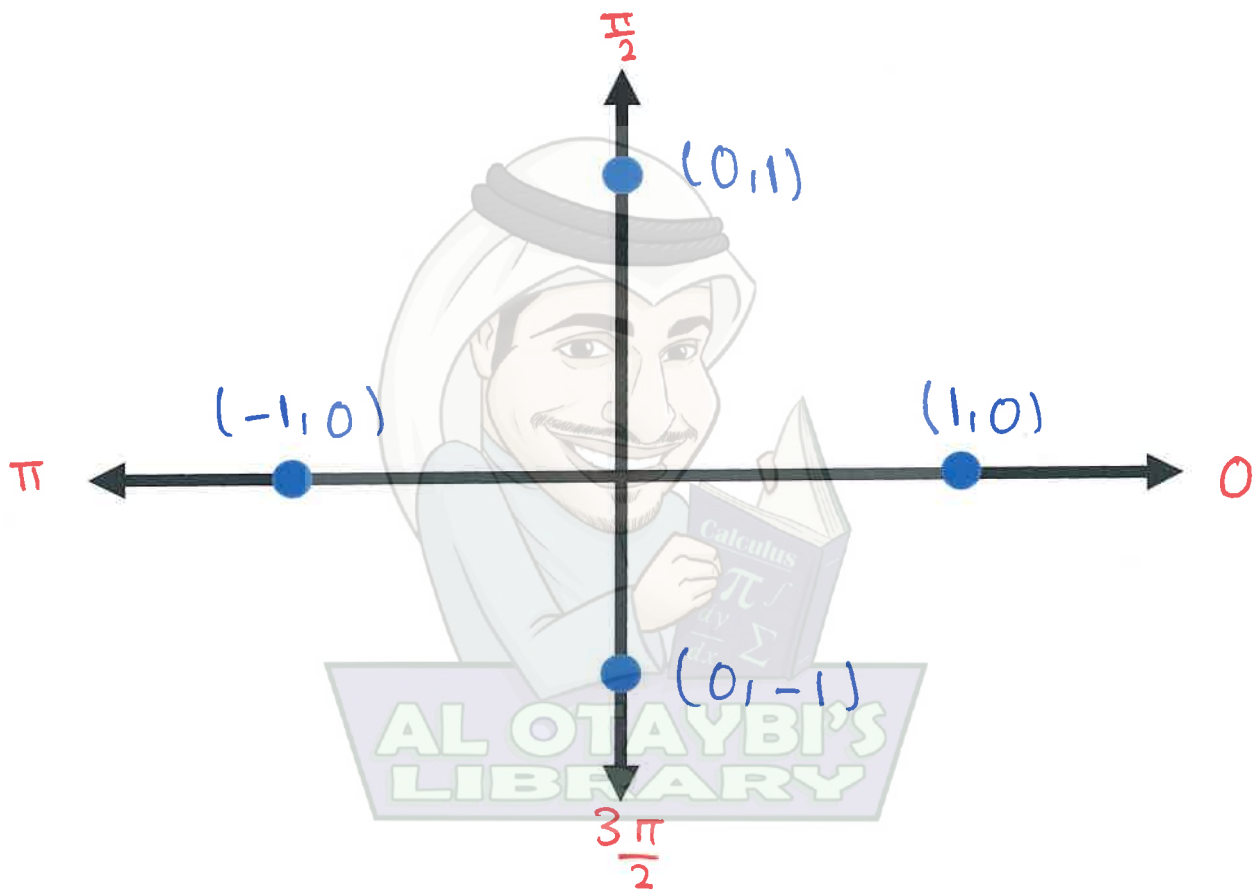
Review for angles
مراجعة على الزوايا



الزوايا القائمة



$$P(x, y) = (\cos\theta, \sin\theta)$$



Find the value of

$$\sin(0) = 0$$

$$\tan(0) = \frac{\sin 0}{\cos 0} = \frac{0}{1} = 0$$

$$\cos(0) = 1$$

$$\sec(\pi) = \frac{1}{\cos \pi} = \frac{1}{-1} = -1$$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

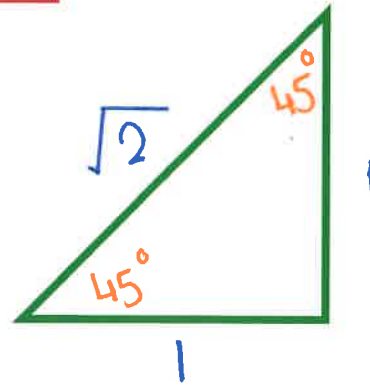
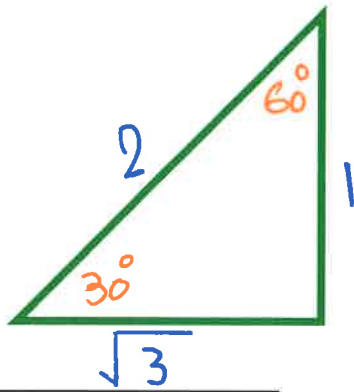
$$\csc\left(\frac{3\pi}{2}\right) = \frac{1}{\sin \frac{3\pi}{2}} = \frac{1}{-1} = -1$$



الزوايا $45^\circ - 60^\circ - 30^\circ$



مثلثات



Find the value of

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

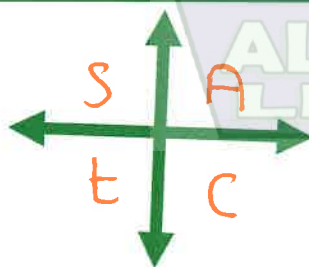
$$\tan\left(\frac{\pi}{4}\right) = \frac{1}{1} = 1$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\sec\left(\frac{\pi}{4}\right) = \frac{1}{\cos\frac{\pi}{4}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$



الزوايا الشاذة



الإشارة



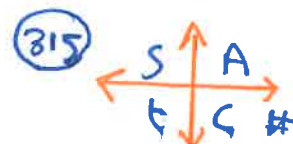
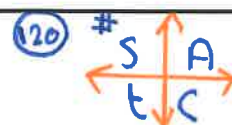
(1) أشوف الربع

(2) الزاوية = $\frac{\pi}{\text{المقام}}$

Find the value of

$$\sin\left(\frac{2\pi}{3}\right) = + \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\sin\left(\frac{7\pi}{4}\right) = - \sin \frac{\pi}{4} = - \frac{1}{\sqrt{2}}$$





سوالف !!



$\sin^{-1}x$

Domain = $[-1, 1]$

Range = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}(1) = \frac{\pi}{2}$

$\sin^{-1}(-1) = -\frac{\pi}{2}$

$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$

$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$

$\sin^{-1}(0) = 0$



$\tan^{-1}x$

Domain = R

Range = $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\tan^{-1}(1) = \frac{\pi}{4}$

$\tan^{-1}(-1) = -\frac{\pi}{4}$

$\tan^{-1}(0) = 0$



$\cos^{-1}x$

Domain = $[-1, 1]$

Range = $[0, \pi]$

$\cos^{-1}(1) = 0$

$\cos^{-1}(-1) = \pi$

$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{3\pi}{3} - \frac{\pi}{3} = \frac{2\pi}{3}$

$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$

$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$\cos^{-1}(0) = \frac{\pi}{2}$



Text book questions



Find the exact value of each expression

$$\cos^{-1}(-1) = \pi$$

$$\sin^{-1}(0.5) = \frac{\pi}{6}$$

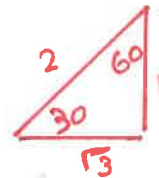
$$\tan^{-1}\sqrt{3} = \frac{\pi}{3}$$

$$\arctan(-1) = -\frac{\pi}{4}$$

$$\arcsin 1 = \frac{\pi}{2}$$

$$\sin^{-1}(-1/\sqrt{2}) = -\frac{\pi}{4}$$

$$\cos^{-1}(\sqrt{3}/2) = \frac{\pi}{6}$$



Tricky questions



Find the exact value of each expression

$$\csc^{-1}\sqrt{2} = \frac{\pi}{4}$$

$$\sec^{-1}2 = \frac{\pi}{3}$$

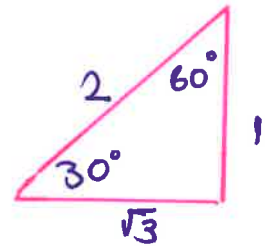
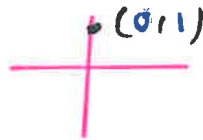
$$\cot^{-1}(-\sqrt{3}) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Evaluate $\sin^{-1}\left(\sqrt{3} \tan \frac{\pi}{6}\right)$

$$= \sin^{-1}\left(\sqrt{3} \frac{1}{\sqrt{3}}\right)$$

$$= \sin^{-1}(1)$$

$$= \frac{\pi}{2}$$

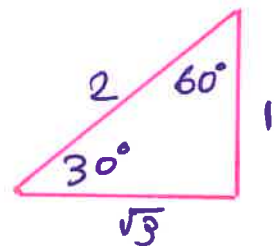


Evaluate $\cos^{-1}\left(2 \cos \frac{\pi}{3}\right)$

$$= \cos^{-1}\left(2 \frac{1}{2}\right)$$

$$= \cos^{-1}(1)$$

$$= 0$$

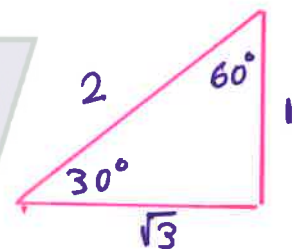


Evaluate $\tan^{-1}\left(\sqrt{3} \tan \frac{\pi}{6}\right)$

$$= \tan^{-1}\left(\sqrt{3} \frac{1}{\sqrt{3}}\right)$$

$$= \tan^{-1}(1)$$

$$= \frac{\pi}{4}$$



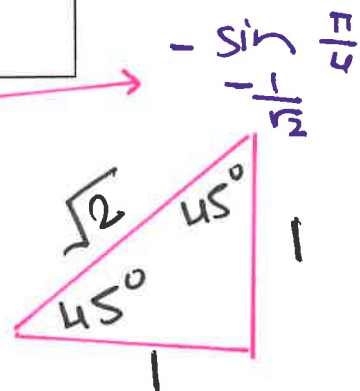
S	A
#t	c

Evaluate $\cos^{-1}\left(\sin \frac{5\pi}{4}\right)$

$$= \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$= \pi - \frac{\pi}{4}$$

$$= \frac{4\pi}{4} - \frac{\pi}{4} = \frac{3\pi}{4}$$





سوالف !!



جرائم !!



$$\sin^{-1}x = (\sin x)^{-1}$$

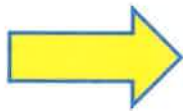
$$\csc^{-1}x = \frac{1}{\sin x}$$

$$\sec^{-1}x = \frac{1}{\cos x}$$

$$\cot^{-1}x = \frac{1}{\tan x}$$

**AL OTAYBI'S
LIBRARY**

$$\sin^{-1}x$$



inverse



$$(\sin x)^{-1}$$



Power = -1



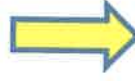
$$\arcsin x = \sin^{-1}x$$



المختصر المفيد



$$\sin^{-1} \sin x = x$$



يجب أن تنتمي
x
الى
 $\text{Range } \sin^{-1}$

$$\sin \sin^{-1} x = x$$



يجب أن تنتمي
x
الى
 $\text{Domain } \sin^{-1}$

True or false (Justify your answer)

$$\sin^{-1} \sin \frac{\pi}{3} = \frac{\pi}{3}$$

$$\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \Rightarrow \therefore \text{True}$$

$$\sin^{-1} \sin \frac{2\pi}{3} = \frac{2\pi}{3}$$

$$\frac{2\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \Rightarrow \therefore \text{False}$$

$$\sin \sin^{-1} 1 = 1$$

$$1 \in [-1, 1] \\ \Rightarrow \therefore \text{True}$$

$$\sin \sin^{-1} 3 = 3$$

$$3 \notin [-1, 1] \\ \Rightarrow \therefore \text{False}$$

Find the exact value of

$$\sin^{-1}\left(\sin\frac{\pi}{4}\right)$$

~~$\sin^{-1}(\sin \frac{\pi}{4}) = \frac{\pi}{4}$~~ $\frac{\pi}{4} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Find the exact value of

$$\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$$

$$\begin{aligned} &= \tan^{-1}(-\sqrt{3}) \\ &= -\frac{\pi}{3} \end{aligned}$$

$\frac{2\pi}{3} \notin (-\frac{\pi}{2}, \frac{\pi}{2})$
 $\tan \frac{2\pi}{3} \rightarrow \begin{array}{c} \# \text{ } \begin{array}{|c|} \hline A \\ \hline T \\ \hline C \\ \hline \end{array} \\ \hline \end{array}$
 $= -\tan \frac{\pi}{3} = -\frac{\sqrt{3}}{1}$

Find the exact value of

$$\tan^{-1}\left(\tan\frac{13\pi}{4}\right)$$

$$\begin{aligned} &= \tan^{-1}(1) \\ &= \frac{\pi}{4} \end{aligned}$$

$\frac{13\pi}{4} \notin (-\frac{\pi}{2}, \frac{\pi}{2})$
 $\tan \frac{13\pi}{4}$
 $= \tan\left(\frac{13\pi}{4} - 2\pi\right)$
 $= \tan\left(\frac{5\pi}{4}\right) = \tan \frac{\pi}{4} = 1$



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Text book question



Find the exact value of

$$\arcsin(\sin(5\pi/4))$$

$$\begin{aligned} &= \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) \\ &= -\frac{\pi}{4} \end{aligned}$$

$\frac{5\pi}{4} \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$
 $\sin \frac{5\pi}{4}$
 $= -\sin \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$
 $\begin{array}{c} \begin{array}{|c|} \hline S \\ \hline T \\ \hline C \\ \hline \end{array} \begin{array}{|c|} \hline A \\ \hline C \\ \hline \end{array} \\ \hline \end{array}$

بعض قوانين الدوال المثلثية



$$\cos^2 x + \sin^2 x = 1$$

$$\div \cos^2 x$$

$$1 + \tan^2 x = \sec^2 x$$

$$\div \sin^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

$$\sin(2x) = 2\sin x \cos x$$

$$\cos(2x) = \begin{cases} \cos^2 x - \sin^2 x \\ 2\cos^2 x - 1 \\ 1 - 2\sin^2 x \end{cases}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

$$\sin(-x) = -\sin(x) \quad , \quad \tan(-x) = -\tan(x)$$
$$\cos(-x) = \cos(x)$$

cos

تبلغ السالب
بزوايتها



(دالة مثلثية بالانفرس) دالة مثلثية

Examples : $\sin(\tan^{-1}x) - \cos(\sec^{-1}x) \dots$



خطوات الحل



1) الدالة المثلثية بالانفرس $y =$

2) أطير دالة الانفرس

3) رسم المثلث



انتبه



إذا كان هناك رقم قبل أو بعد الانفرس

إذا كانت هناك أكثر من دالة انفرس



يتم اهماله



أكثر من مثلث

Find the exact value of $\sin\left(\cos^{-1}\left(\frac{4}{5}\right)\right)$

Mar 2019

Let $y = \cos^{-1}\left(\frac{4}{5}\right)$

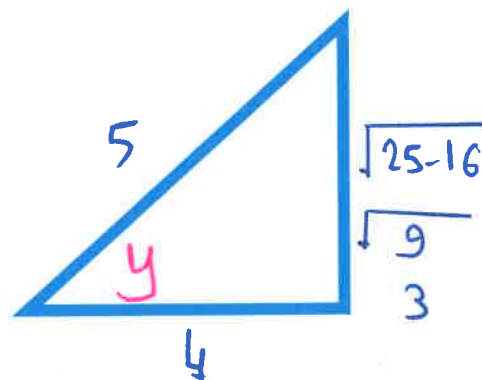
$\cos y = \cos \cos^{-1}\left(\frac{4}{5}\right)$

$\cos y = \frac{4}{5}$

$\Rightarrow \sin\left(\cos^{-1}\left(\frac{4}{5}\right)\right)$

$\sin y$

$= \frac{3}{5}$



Find the exact value of $\sin\left(\sec^{-1}\left(\frac{5}{3}\right)\right)$

Oct 2018

Let $y = \sec^{-1}\left(\frac{5}{3}\right)$

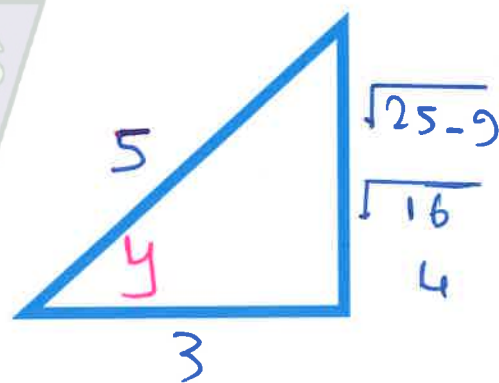
$\sec y = \sec \sec^{-1}\left(\frac{5}{3}\right)$

$\sec y = \frac{5}{3}$

$\Rightarrow \sin\left(\sec^{-1}\left(\frac{5}{3}\right)\right)$

$= \sin y$

$= \frac{4}{5}$





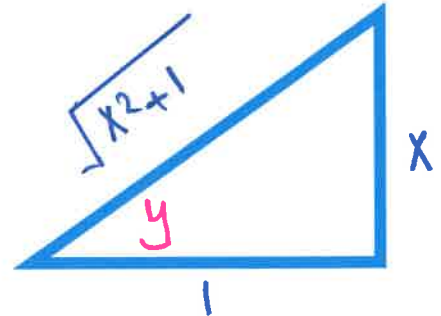
Text book questions



Simplify the expression

$$\sin(\tan^{-1}x)$$

$$\begin{aligned}\text{Let } y &= \tan^{-1}x \\ \tan y &= \tan \tan^{-1}x \\ \tan y &= x\end{aligned}$$

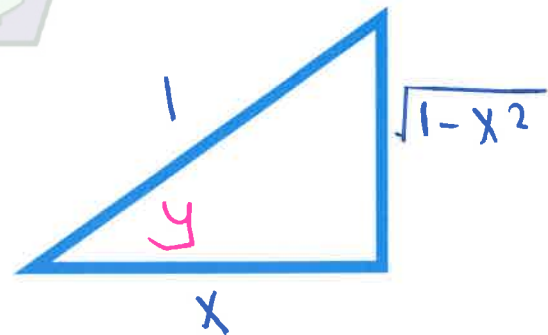


$$\begin{aligned}\Rightarrow \sin(\tan^{-1}x) \\ &= \sin y \\ &= \frac{x}{\sqrt{x^2 + 1}}\end{aligned}$$

Simplify the expression

$$\sin(2\arccos x)$$

$$\begin{aligned}\text{Let } y &= \cos^{-1}x \\ \cos y &= \cos \cos^{-1}x \\ \cos y &= x\end{aligned}$$



$$\begin{aligned}\Rightarrow \sin(2\cos^{-1}x) \\ &= \sin(2y) \\ &= 2 \sin y \cdot \cos y \\ &= 2 \cdot \frac{\sqrt{1-x^2}}{1} \cdot \frac{x}{1} = 2x \sqrt{1-x^2}\end{aligned}$$



Text book question



Find the exact value of

$$\cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right)$$

Let $y = \sin^{-1}\left(\frac{5}{13}\right)$

$$\sin y = \frac{5}{13}$$

$$\Rightarrow \cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right)$$

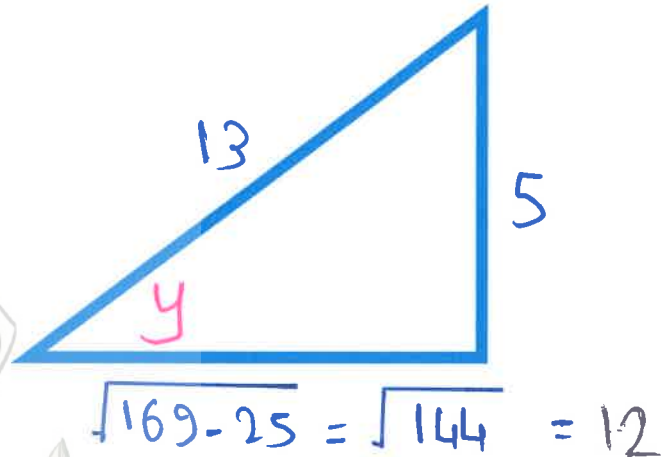
$$= \cos(2y)$$

$$= \cos^2 y - \sin^2 y$$

$$= \left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2$$

$$= \frac{144}{169} - \frac{25}{169}$$

$$= \frac{119}{169}$$





Text book question



Prove that

$$\cos(\sin^{-1}x) = \sqrt{1-x^2}$$

$$\text{L.H.S} = \cos(\sin^{-1}x)$$

$$\text{Let } y = \sin^{-1}x$$

$$\sin y = \cancel{\sin} \sin^{-1}x$$

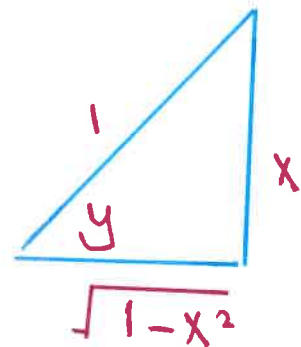
$$\sin y = x$$

$$\Rightarrow \cos(\sin^{-1}x)$$

$$= \cos(y)$$

$$= \frac{\sqrt{1-x^2}}{1}$$

$$= \sqrt{1-x^2}$$



AL OTAYBI'S
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$$= \text{R.H.S}$$

Find the exact value of

$$\sec\left(\sin^{-1}\frac{1}{3}\right) - \sin(\sec^{-1}3)$$

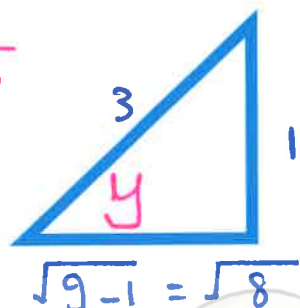
Let $y = \sin^{-1}\frac{1}{3}$

$$\sin y = \frac{1}{3}$$

$$\Rightarrow \sec\left(\sin^{-1}\frac{1}{3}\right)$$

$$\sec y$$

$$= \frac{3}{\sqrt{8}}$$



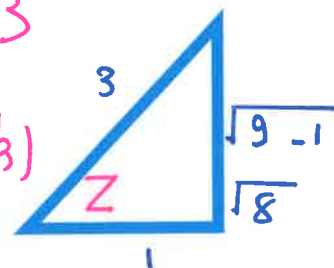
Let $z = \sec^{-1}3$

$$\sec z = 3$$

$$\Rightarrow \sin(\sec^{-1}3)$$

$$= \sin z$$

$$= \frac{\sqrt{8}}{3}$$



Homework



Find the exact value of the expression

$$\cos\left(\sin^{-1}\left(\frac{1}{3}\right) + \cos^{-1}\left(\frac{2}{3}\right)\right)$$

$$\text{Final Answer : } \frac{2\sqrt{8} - \sqrt{5}}{9}$$

a) Show that $\sin(\tan^{-1}x) = \frac{x}{\sqrt{1+x^2}}$

b) Find the exact value of $\sin\left(\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

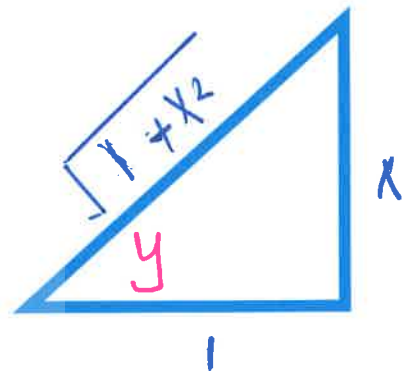
Nov 2016

a) L.H.S = $\sin(\tan^{-1}x)$

Let $y = \tan^{-1}x$

$\tan y = \tan \tan^{-1}x$

$\tan y = x$



$\Rightarrow \sin(\tan^{-1}x) = \sin y = \frac{x}{\sqrt{1+x^2}} = \text{R.H.S}$

b) $\sin\left(\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right) = \frac{\frac{1}{\sqrt{3}}}{\sqrt{1+\left(\frac{1}{\sqrt{3}}\right)^2}}$
 $= \frac{\frac{1}{\sqrt{3}}}{\sqrt{1+\frac{1}{3}}} = \frac{\frac{1}{\sqrt{3}}}{\sqrt{\frac{4}{3}}} = \frac{\frac{1}{\sqrt{3}}}{\frac{2}{\sqrt{3}}} = \frac{1}{2}$



Homework



Find the exact value of the expression

$\tan\left(\sin^{-1}\left(\frac{2}{3}\right)\right)$

July 2017

Final Answer : $\frac{2}{\sqrt{5}}$

Let $f(x) = \sin^{-1}(2x - 1)$

a) Find the domain of $f(x)$

b) Find $f^{-1}(x)$

c) Find the range of $f(x)$

a) $-1 \leq 2x - 1 \leq 1$
 $0 \leq 2x \leq 2$
 $\div 2$
 $0 \leq x \leq 1$
 $\Rightarrow \therefore D_f = [0, 1]$

b) $y = \sin^{-1}(2x - 1)$
 $\sin y = \sin \sin^{-1}(2x - 1)$
 $\sin y = 2x - 1$
 $\sin y + 1 = 2x$
 $\div 2$
 $x = \frac{1}{2} [\sin y + 1]$
 $\Rightarrow \therefore f^{-1}(x) = \frac{1}{2} [\sin x + 1]$

c) Domain of $f^{-1} = \mathbb{R}$

$\therefore$ Range of $f(x) = \mathbb{R}$



Tricky Question !!



Solve the equation

طلع فرقة
x

$$\sec(\tan^{-1}x) = \sqrt{2x}$$

$$\sec(\tan^{-1}x)$$

$$y = \tan^{-1}x$$

$$\tan y = \tan \tan^{-1}x$$

$$\tan y = x$$

$$\sec(\tan^{-1}x)$$

$$\sec y = \frac{\sqrt{1+x^2}}{1} = \sqrt{1+x^2}$$

$$\Rightarrow \sec(\tan^{-1}x) = \sqrt{2x}$$

$$\sqrt{1+x^2} = \sqrt{2x}$$

$$1+x^2 = 2x$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)(x-1) = 0$$

$$x-1 = 0$$

$$x=1$$

$$x-1 = 0$$

$$x=1$$

$$\Rightarrow \therefore x=1$$